

Şirag Erkol

CONTACT INFORMATION	Northwestern University Center for Science of Science and Innovation 1800 Sherman Ave, Evanston, IL 60201	(812) 558-8938 siragerkol@gmail.com siragerkol.github.io
ACADEMIC POSITIONS	Northwestern University , Evanston, IL <i>Research Assistant Professor</i>	July 2025 – present
	Northwestern University , Evanston, IL <i>Postdoctoral Fellow</i>	July 2023 – June 2025
	Indiana University , Bloomington, IN <i>Research & Teaching Assistant</i>	August 2017 – May 2023
	Boğaziçi University , Istanbul, Turkey <i>Research & Teaching Assistant</i>	October 2014 – August 2017
EDUCATION	Indiana University , Bloomington, IN <i>Ph.D. in Informatics, Complex Networks and Systems track</i> • Advisor: <i>Assoc. Prof. Filippo Radicchi</i> • Thesis: <i>Identifying Influential Spreaders in Complex Networks</i> • Minor: Computer Science <i>M.S. in Informatics</i>	August 2017 – May 2023
	Boğaziçi University , Istanbul, Turkey <i>M.S. in Industrial Engineering</i> • Advisor: <i>Assoc. Prof. Gönenç Yücel</i> • Thesis: <i>Influence Maximization Based on Partial Network Structure Information: A Comparative Analysis on Seed Selection Heuristics</i> <i>B.S. in Industrial Engineering</i>	September 2014 – August 2016 September 2010 – June 2014
WORK EXPERIENCE	Pacific Northwest National Laboratory , Richland, WA <i>Ph.D. Intern</i>	June 2021 – August 2021
HONORS & AWARDS	IU Luddy Summer Research Award (\$4400)	2022
PUBLICATIONS	Published: Symmetry breaking in optimal transport networks S. Patwardhan, M. Barthelemy, Ş. Erkol , S. Fortunato, F. Radicchi, <i>Nature Communications</i> , 15, 3758 (2024) [DOI] [arXiv] Consistency pays off in science Ş. Erkol , S. Sikdar, F. Radicchi, and S. Fortunato, <i>Quantitative Science Studies</i> , 4, 491 (2023) [DOI] [arXiv] [GitHub] [Website] Effective submodularity of influence maximization on temporal networks Ş. Erkol , D. Mazzilli and F. Radicchi, <i>Physical Review E</i> , 106, 034301 (2022) [DOI] [arXiv]	

Who is the best coach of all time? A network-based assessment of the career performance of professional sports coaches

Ş. Erkol and F. Radicchi, *Journal of Complex Networks*, 9, cnab012 (2021)

[DOI] [arXiv] [Website]

Influence maximization on temporal networks

Ş. Erkol, D. Mazzilli and F. Radicchi, *Physical Review E*, 102, 042307 (2020)

[DOI] [arXiv]

Systematic comparison between methods for the detection of influential spreaders in complex networks

Ş. Erkol, C. Castellano and F. Radicchi, *Scientific Reports*, 9, 15095 (2019)

[DOI] [arXiv]

Influence maximization in noisy networks

Ş. Erkol, A. Fageeh and F. Radicchi, *EPL*, 123, 58007 (2018)

[DOI] [arXiv]

Influence Maximization Based on Partial Network Structure Information: A Comparative Analysis on Seed Selection Heuristics

Ş. Erkol and G. Yücel, *International Journal of Modern Physics C*, 28, 1750122 (2017)

[DOI]

PRESENTATIONS

Talks:

Second acts in scientific and creative careers

June 2026

NetSci 2026, Boston, MA, USA

Effective submodularity of influence maximization on temporal networks

July 2022

NetSci 2022, Online

Influence maximization on temporal networks

July 2021

Networks2021, Online

Influence maximization on temporal networks

March 2021

University of Limerick, Ireland, Online

Posters:

Consistency pays off in science

June 2023

ICSSI 2023, Evanston, IL, USA

Systematic comparison between methods for the detection of influential spreaders in complex networks

September 2020

NetSci 2020, Online

Influence maximization in noisy networks

May 2019

NetSci 2019, Burlington, VT, USA

COMMUNITY SERVICE

Journal Reviewer: Applied Mathematics and Computation; Chaos, Solitons & Fractals; EPJ Data Science; Expert Systems with Applications; IEEE Transactions on Network Science and Engineering; Journal of Complex Networks; Journal of Computational Science; Pattern Recognition; Physical Review E; Royal Society Open Science; Science Advances

Conference Reviewer: ISSI (2023); ICSSI (2024); IC2S2 (2026)

Journal Subreviewer: PLoS One, PNAS

COMPUTER SKILLS

Frequent user of Python, L^AT_EX

Experience in Django, NetLogo, HTML, CSS

Familiar with MySQL, C, R, Git, Stella

LANGUAGES English (fluent), Turkish (native), Armenian (native), French (beginner), Spanish (beginner)

Last updated: June 2026